

Autodesk® Scaleform®

HUD

Scaleform 4.3 HUD

Flash UI

C++

AAA

Nate Mitchell, Prasad Silva

2.0

2010 7 30

Copyright Notice

Autodesk® Scaleform® 4.3

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

HUD

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

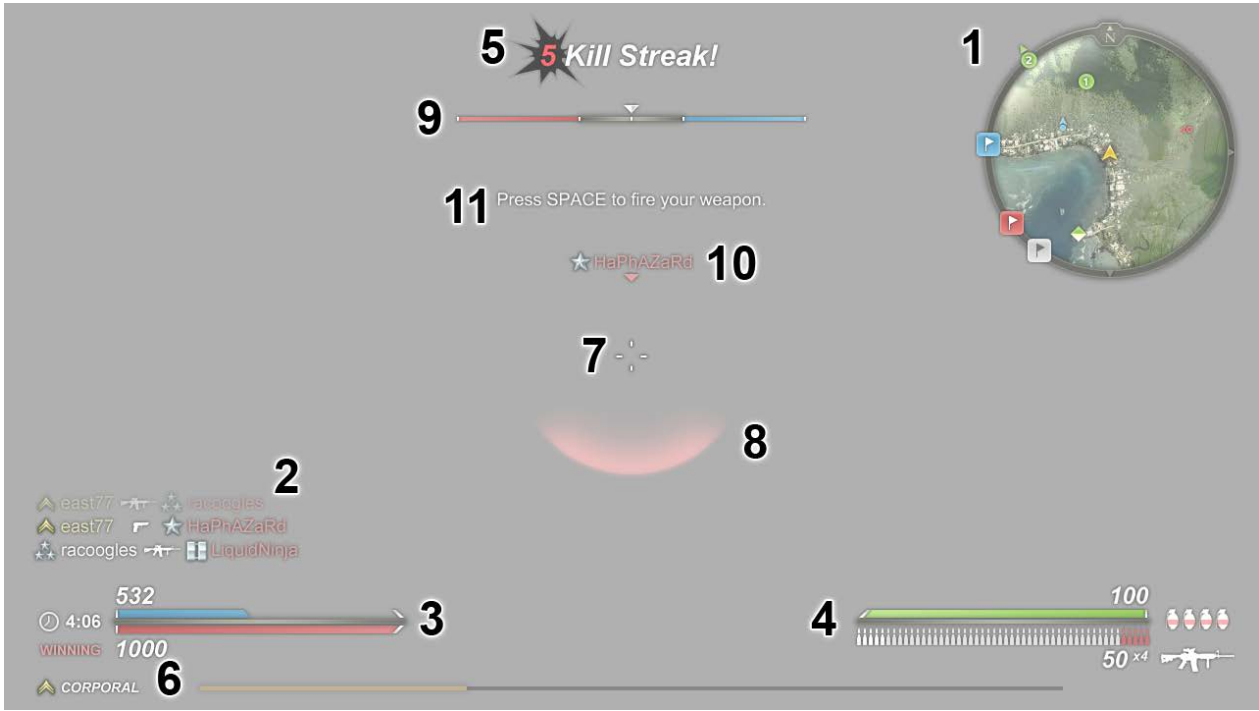
(301) 446-3199

1		1
2		2
2.1	(Build)	2
2.2		3
2.3		3
2.3.1	(Windows).....	3
2.3.2	(Xbox/PS3)	4
3		5
3.1	C++.....	5
3.1.1		5
3.1.2	HUD/	5
3.1.3		6
3.2	Flash	6
3.2.1	HUDKit fla	6
3.2.2	Minimap fla	7
4 HUD		9
4.1		9
4.2	Player Stats Ammo	11
4.3		15
4.4		16
4.5		17
4.6		20
4.7		20
4.8		21
4.9		22

4.10		24
5		28
5.1		28
6		30
6.1		30
6.2		31

1 引言

Scaleform HUD() AAA
HUD
Scaleform® Direct Access API (FPS) UI



1 HUD

- HUD UI :
- 1. (Minimap)
 - 2.
 - 3.
 - 4. / /
 - 5.
 - 6.
 - 7.
 - 8.
 - 9.
 - 10. /
 - 11.

FPS

2 概述

2.1 檔位置和生成 (Build) 注釋

- :
- *Apps\Kits\HUD* - HUD C++
 - *Bin\Data\AS2 or AS3\Kits\HUD* - Flash ActionScript (AS2 ActionScript 2 Flash 8 , AS3 ActionScript 3 Flash 10) 。
 - *Projects\Win32 \{Msvc80 or Msvc90}\Kits\HUD* – Windows Visual Studio 2005/2008
 - *Projects\Xbox360\{Msvc80 or Msvc90}\Kits\HUD* – Xbox 360 Visual Studio 2005/2008
 - *Projects\Common\HudKit.mk* – HUD PS3

Bin\Kits\HUD (Windows) HUDKit.exe
Scaleform SDK

Windows , *Projects\Win32\Msvc80\Kits(Msvc90\Kits)* Scaleform 4.3 Kits.sln
Debugging() “Working ”

Bin\Data\AS2\Kits\HUD or *Bin\Data\AS3\Kits\HUD*

Xbox 360 , *Projects\Xbox360\{Msvc80 Msvc90}\Kits\HUD* Scaleform 4.3 Kits.sln
(*Bin\Data\AS2 or As3\Kits\HUD*)
Xbox 360

PS3, Scaleform make , HUD
(HUD Kit) PS3 , SN Systems .
Bin\PS3 :
app_home/ Directory: {Local Scaleform Directory}*Bin\Data\AS2 or AS3\Kits\HUD*

2.2 演示使用

HUD, , :

, () 3

1 1 500 , (15)

, 4 ,

(Power-up)

2.3 控制方案

2.3.1 鍵盤 (Windows)

W, A, S, D	WS	A, D
SPACE		
R		
G		,
ESC	(Text Field)	,
B	AI	
1, 2	• 1 • 2	

2.3.2 (Xbox/PS3)

	W, S	A, D
X /		
		,
B /		,
Y /		

3 構

HUD Flash UI HUD C++ C++ ,
HUD

3.1 C++

C++ HUD MVC ,
Flash UI Fx HUD "(View)"
HUD HUD HUD
HUD HUD HUD
Scaleform , Scaleform Direct Access API
UI

C++ (Apps\Kits\HUD\)

3.1.1

- **HUDKitDemo.cpp** – Scaleform Player C++ SWF
- **HUDKitDemoConfig.** – FxPlayerConfig.h HUDKitDemo Windows

3.1.2 HUD/

- **FxHUDKit.h** – HUD HUD
- **FxHUDView.h/.cpp** – HUD , (Billboard)
- **FxMinimap.h** –
- **FxMinimapView.h/.cpp** –

3.1.3 戲

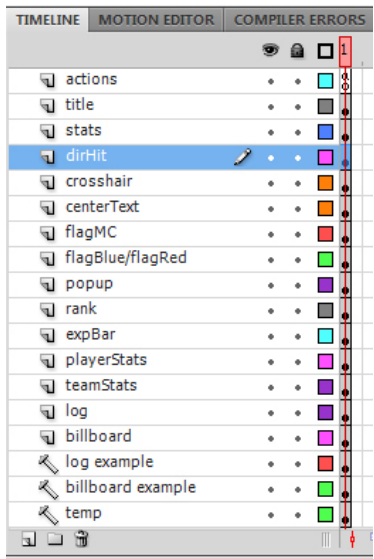
- HUDAdapter.h/.cpp – , HUD ,
- HUDEntityController.h/.cpp – , HUD ,
- HUDSimulation.h/.cpp – UI

3.2 Flash

HUD Flash :HUDKit.fla Minimap.fla FLA HUD
UI , Scaleform UI
 ,
 ,
 ,
Scaleform , C++ ActionScript Flash Flash , HUD
Classic Tweens Flash Gfx::Value::GotoAndPlay()
C++

3.2.1 HUDKit.fla

Bin\Data\AS2 or AS3\Kits\HUD HUDKit.fla SWF HUDKitDemo
Flash UI , HUDKit.fla



2 HUDKit.fla

HUDKitDemo Frame 1(1) External Interface HUDKit MovieClip:

```
ExternalInterface.call("registerHUDView", this);
```

```
MovieClip( ) ,C++ MovieClip C++ HUDView
```

```
"HUD " MovieClips Scaleform C++
```

3.2.2 Minimap.fla

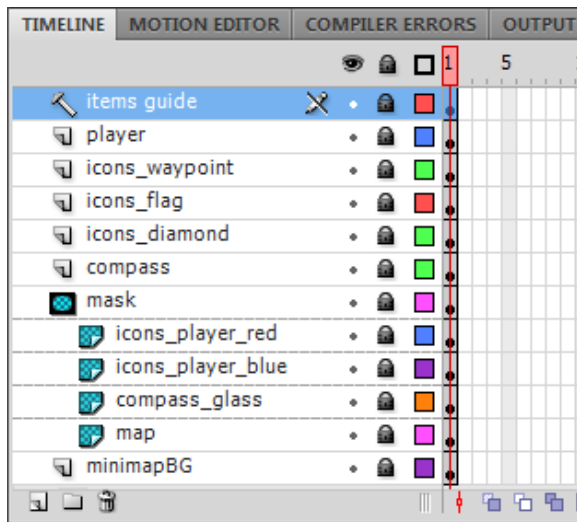
3 Minimap

```
Minimap.fla 'Minimap' , , ,
3D
, (" " ) ,
:
```

- ()
- ()
- ((Flag) : , ,)
- ()

```
Direct Access API
C++
/
MovieClips (Stage) MovieClips
```

'Minimap'



4 'minimap'

- **items guide** / (Guide Layer) SWF
- **player**
- **icons_waypoint**
- **icons_flag**
- **icons_diamond**
- **compass** (" ")
- **mask** (Mask Layer)
 - o **icons_player_red**
 - o **icons_player_blue**
 - o **compass_glass**
 - o **map**
- **minimapBG**

'Minimap' com.scaleform.Minimap (Bin\Data\AS2 or AS3\Kits\HUD\com\scaleform))

HUDKitDemo

4 HUD 視圖

C++ HUD HUDKit.swf :

- FxHUDView – HUD(HUDKit.swf) (Core Manager) MovieClips
- BillboardCache – / (,
- FxHUDLog – HUD FxHUDMessages
- FxHUDMessage – ,

,FxHUDView MC(MovieClip)

,FxHUDView UpdateView() MovieClip

HUDKit fla UI

FxHUDView::UpdateView()
 :UpdateTeamStats() UpdatePlayerEXP() Update PlayerStats() UpdateEvents() UpdateTutorial()
 UpdateBillboards() , UI

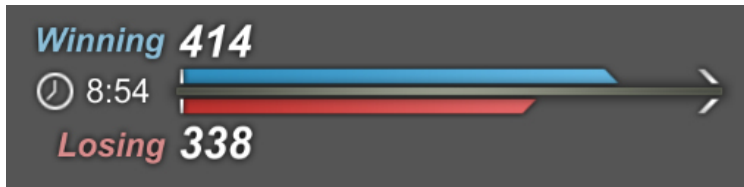
, UpdateView() ,

Scaleform Direct Access API, Flash
 ,Gfx::Value Scaleform Flash
 Gfx::Value::SetDisplayInfo() Gfx::Value::GetDisplayInfo() Gfx::Value::SetText() HUD

,

4.1 回合統計資料和記分牌

teamStats MovieClip(HUDKit fla 1 (Scene 1) teamStats)
 (redWinning, blueWinning) (scoreRed, scoreBlue)
 (teamRed, teamBlue), (roundTime)



5 teamStats

FxHUDView::UpdateTeamStats()

teamStats

teamStats MovieClip

```
void FxHUDView::UpdateTeamStats(FxHUDEnvironment *penv)
{
    // Retrieve the scores from the simulation.
    unsigned scoreBlue = unsigned(penv->GetHUDBlueTeamScore());
    unsigned scoreRed = unsigned(penv->GetHUDRedTeamScore());

    String text;
    Value tf;
    Value::DisplayInfo info;

    // We won't update any element who has not changed since the last update
    // to avoid unnecessary updates. To do so, we compare the previous
    // simulation State to the latest simulation data. If the two are different,
    // update the UI element in the HUD View.
    if (State.ScoreRed != scoreRed)
    {
        Format(text, "{0}", scoreRed);
        ScoreRedMC.SetText(text); Update the red team score text.

        info.SetScale((scoreRed / (Double)MaxScore) * 100, 100);
        TeamRedMC.SetDisplayInfo(info); // Update and scale the red team score
                                         bar movieClip.
    }

    if (State.ScoreBlue != scoreBlue)
    {
        Format(text, "{0}", scoreBlue);
        ScoreBlueMC.SetText(text); // Update the blue team score text.

        info.SetScale((scoreBlue / (Double)MaxScore) * 100, 100);
        TeamBlueMC.SetDisplayInfo(info); // Update and scale the blue team
                                         score bar movieClip.
    }

    if (State.ScoreRed != scoreRed || State.ScoreBlue != scoreBlue)
    {
        // Update the "Winning" and "Losing" text fields for both teams
        // if the score has changed.
        if (scoreBlue > scoreRed)
```

```

        {
            RedWinningMC.SetText(LosingText);
            BlueWinningMC.SetText(WinningText);
        }
        else
        {
            RedWinningMC.SetText(WinningText);
            BlueWinningMC.SetText(LosingText);
        }
    }

    // Update the roundtime clock
    float secondsLeft = penv->GetSecondsLeftInRound();
    unsigned sec = ((unsigned)secondsLeft % 60);
    if (sec != State.Sec)
    {
        unsigned min = ((unsigned)secondsLeft / 60);
        String timeLeft;
        Format(timeLeft, "{0}:{1:0.2}", min, sec);
        RoundTimeMC.SetText(timeLeft);
        State.Sec = sec;
    }
}

Scaleform::String , Gfx::Value::SetText() scoreRed scoreBlue
teamRed teamBlue X Gfx::Value::SetDisplayInfo() 0%

,

blueWinning/redWinning ,
Gfx::Value::SetText() HUDView "Winning" "Losing"
textField , UpdateView()
, Gfx::Value::SetText()
roundTime textField

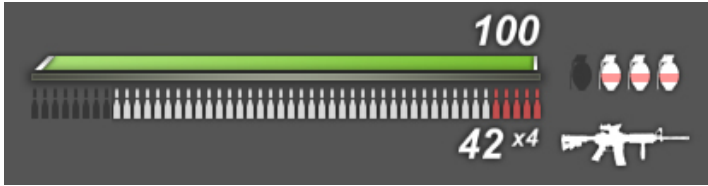
```

4.2 Player Stats Ammo

```

playerStats ( HUDKit fla 1 playerStats ) (healthN,
health) (weapon) HUD (reload)

```

6 playerStats

, clipN ammoN ammo ammoBG grenade grenadeBG

clipN ammoN

ammo/healthVehicle :ammo ammoBG,

51 Keyframes(),

Gfx::Value::GotoAndStop()

- *ammo*
- *ammoBG*

grenade/armor ammo , :grenade grenadeBG,

5

1 4 , 5 [Gfx::Value::GotoAndStop\(\)](#)

- *grenade*
- *grenadeBG*

playerStats

```
void FxHUDView::UpdatePlayerStats(FxHUDEnvironment *penv)
{
```

```
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
```

```
    // Load latest player info.
```

```
    unsigned ammoInClip = pplayer->GetNumAmmoInClip();
```

```
    unsigned ammoClips = pplayer->GetNumClips();
```

```
    unsigned clipSize = pplayer->GetMaxAmmoInClip();
```

```
    unsigned grenades = pplayer->GetNumGrenades();
```

```
    unsigned ammoFrames = 51;
```

```
    unsigned grenadeFrames = 5;
```

```
    String text;
```

```

if(ppplayer->GetHealth() != State.Health)
{
    unsigned health = UInt(ppplayer->GetHealth() * 100);
    Format(text, "{0}", health);
    HealthNMC.SetText(text); // Update the health text field.

    Value::DisplayInfo info;
    info.SetScale(Double(health), 100);
    HealthMC.SetDisplayInfo(info); // Update and scale the health bar.
}

if(ammoInClip != State.AmmoInClip)
{
    Format(text, "{0}", ammoInClip);
    AmmoNMC.SetText(text); // Update the ammo text field.

    // The ammo is divided into two parts: the white bullet icons (AmmoMC)
    // and their respective backgrounds (AmmoBGMC).

    // To update the number of bullets currently in the clip, we use
    // GotoAndStop with AmmoMC and select the frame with the proper number
    // of white bullet icons. There are 51 frames in AmmoMC (defined above
    // in ammoFrames) and the first frame displays every bullet icon.

    // To display X number of bullets, we subtract X from the total number
    // of frames in AmmoMC (51) and GotoAndStop on the result.
    AmmoMC.GotoAndStop(ammoFrames - ammoInClip);
}

if(ammoClips != State.AmmoClips)
{
    Format(text, "x{0}", ammoClips);
    ClipNMC.SetText(text); // Update the remaining ammo clips text field.
}

if(grenades != State.Grenades)
{
    // Grenades are setup similar to Ammo. The first frame for the
    // Grenades movieClip shows 4 white grenade icons. The second frame
    // shows 3.

    // To display X number of grenades, we subtract X from the total number
    // of frames in GrenadeMC (5) and GotoAndStop on the result.
    // Note: GrenadeMC actually contains 10 frames but 6-10 are unused by
    // this implementation.
    GrenadeMC.GotoAndStop(grenadeFrames - grenades);
}

```

```

if (pplayer->GetWeaponType() != State.weaponType)
{
    Change the weapon icon if the player has changed weapons.
    unsigned weaponFrame = GetWeaponFrame(pplayer->GetWeaponType());
    WeaponMC.GotoAndStop(weaponFrame);

    // Update the ammo background icons based on clipSize.
    if (clipSize != State.AmmoClipSize)
        AmmoBGMC.GotoAndStop(ammoFrames - clipSize);
}

// If the ammo in clip is less than 6 bullets, play the "Reload" indicator
// movieClip.
if (ammoInClip <= 5)
{
    // ReloadMC will play until we GotoAndStop on Frame 1, so no need to
    // start the animation more than once.
    if (!bReloadMCVisible)
    {
        bReloadMCVisible = true;
        ReloadMC.GotoAndPlay("on"); // Will cycle back to "on" frame when
                                    it reaches the end and play again.
    }
}

// If the reload indicator is displayed but there are more than six bullets
// hide it and stop the animation by playing frame 1 of the movieClip.
else if (bReloadMCVisible)
{
    ReloadMC.GotoAndStop("1"); // Frame 1 contains stop();
    bReloadMCVisible = false;
}
}

ammo , 1 50 , 51 0 ,
ammoMC.GotoAndStop(ammoFrames - ammoInClip) , HUD

ammoBG, ammoBGMC.GotoAndStop() X ,X
, (Machine Gun) 50
,ammoBG.GotoAndStop(1) 1 50
FxHUDView::UpdateView() MovieClip

```

4.3 旗標捕獲指示器

```
flagMC ( HUDKit fla 1 ) flagMC flag ,
: (arrow), CaptureState( )
, , ,
```



7 flagMC

```
flagMC Classic Tween Flash flagMC 4
, 4 :Hidden( ) Visible( ) Fade-In( ) FadeOut( )
GotoAndStop() Keyframe / (1 5 "On" "Off" )
FxHUDView::UpdateEvents() flagMC arrow( )
```

```
// Flag capture indicators and reticule behavior.
void FxHUDView::UpdateEvents(FxHUDEnvironment *penv)
{
    Value::DisplayInfo info;

    if (penv->IsHUDPlayerCapturingObjective())
    {
        if(!bFlagMCVisible)
        {
            SetVisible(&FlagMC, true); // Set the Flag MovieClip's visibility
                                     // to true
            FlagMC.GotoAndPlay("on"); // Play the fade-in animation once.
            bFlagMCVisible = true;
        }

        // Shift the x-coordinate of the Flag Arrow to show the capture state
        // of the flag the player is currently capturing.
        info.SetX(penv->GetHUDCaptureObjectiveState() * 177);
        FlagArrowMC.SetDisplayInfo(info);
        info.Clear();
    }
    else if (bFlagMCVisible & !penv->IsHUDPlayerCapturingObjective())
    {
        // If the flag indicator is still visible and the player is
        // no longer capturing an objective, play the fade-out animation
        FlagMC.GotoAndPlay("off");
    }
}
```

```

        bFlagMCVisible = false;
    }
}

flagMC.GotoAndPlay("on") ,flagMC , Visible( )
flagMC.GotoAndPlay("off") ,flagMC 11-20, 20 ,
1 Flash ; ActionScript C++ 1 stop(); ,
, 1 , Alpha
0

arrow Gfx::Value::SetDisplayInfo\(\) Gfx::Value::DisplayInfo,
CaptureState X

```

4.4 武器十字線

```

Reticule( ) ( HUDKit fla 1 ) reticule :left
right bottom top,
PlayerFire ,

```



8 reticule

```

PlayerFire :FxHUDView::UpdateEvents() FxHUDView::OnEvent()
,ReticuleHeat ReticuleFiringOffset FxHUDEvent

// If the player is firing his/her weapon, update the reticule's heat.
// The reticule elements are updated in FxHUDView::UpdateEvents() based on
// the ReticuleHeat and the ReticuleFiringOffset.
case FxHUDEvent::EVT_PlayerFire:
{
    if (ReticuleHeat < 8)
        ReticuleHeat += 2.5f;

    ReticuleFiringOffset = 2;
}

```

```

FxHUDView::UpdateEvents()
    , SetDisplayInfo()
0 X Y , SetDisplayInfo()
Gfx::Value::DisplayInfo , DisplayInfo
Gfx::Value::SetDisplayInfo() DisplayInfo.Clear() DisplayInfo

```

```

// Shift the XY-coordinates of the reticule elements based on the
// firing duration and rate of fire.

```

```

if (ReticuleHeat > 0 || ReticuleFiringOffset > 0){
    if (ReticuleHeat > 1.0f)
        ReticuleHeat -= .02f;

    ReticuleFiringOffset -= .02f;
    if (ReticuleFiringOffset < 0)
        ReticuleFiringOffset = 0;

    info.SetY((-6) - (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Top.SetDisplayInfo(info);
    info.Clear();

    info.SetX((6) + (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Right.SetDisplayInfo(info);
    info.Clear();

    info.SetX((-6) - (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Left.SetDisplayInfo(info);
    info.Clear();

    info.SetY((6) + (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Bottom.SetDisplayInfo(info);
    info.Clear();
}

```

4.5 定向命中指示器

```

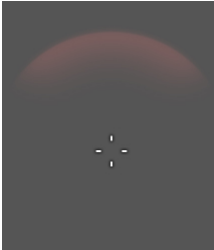
dirHit ( HUDKit fla 1 dirHit ) dirHit
:tl l bl b br r tr t

```

Alpha , :

a)

b)



9 dirHit

Classic Tween Alpha , ~1 Flash
Gfx::Value::GotoAndPlay("on"),

```
void FxHUDView::OnEvent( FxHUDEvent* pevent )
{
    switch ( pevent->GetType() )
    {
        // If the Damage Event (Player was hit) is fired, show the appropriate
        // directional hit indicator.
        // We can use GotoAndPlay("on") to play the fade-in animation. It will
        // fade out on its own after ~1 second (this animation is setup on the
        // Flash timeline)
        case FxHUDEvent::EVT_Damage:
        {
            FxHUDDamageEvent* peventDamage = (FxHUDDamageEvent*)pevent;
            float dir = peventDamage->GetDirection();
            if (dir > 0)
            {
                if (dir > 90)
                {
                    if (dir > 135)
                        DirMC_B.GotoAndPlay("on");
                    else
                        DirMC_BR.GotoAndPlay("on");
                }
                else
                {
                    if (dir > 45)
                        DirMC_TR.GotoAndPlay("on");
                    else
                        DirMC_T.GotoAndPlay("on");
                }
            }
        }
    }
}
```

```

    }
    else
    {
        if (dir < -90)
        {
            if (dir < -135)
                DirMC_B.GotoAndPlay("on");
            else
                DirMC_BL.GotoAndPlay("on");
        }
        else
        {
            if (dir < -45)
                DirMC_L.GotoAndPlay("on");
            else
                DirMC_TL.GotoAndPlay("on");
        }
    }
    break;
}
}
}

```


4.6 等級和經驗顯示

```
expBar ( HUDKit.fla Scene 1 expBar ) ,
expBar exp MovieClip,

rank ( HUDKit.fla Scene 1 rank ) rank 18 Keyframe,
Keyframe
```



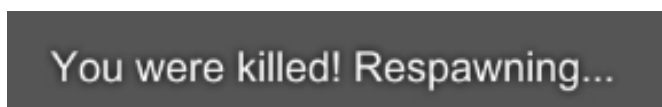
10 expBar rank

```
void FxHUDView::UpdatePlayerEXP(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
    if (pplayer->GetLevelXP() != State.Exp)
    {
        Value::DisplayInfo info;
        // Update the rank icon. Each rank icon is on a different frame of
        // the rank movieClip.
        RankMC.GotoAndStop(UInt(pplayer->GetRank() + 1));
        info.SetScale(pplayer->GetLevelXP() * 100, 100); // 伸縮 EXP 進度條
                                                         // 電影剪輯。
        ExpBarMC.SetDisplayInfo(info);
    }
}
```

```
,exp rank,
GFx::Value::GotoAndStop() X, X + 1( 0)
```

4.7 文本通知

```
centerTextMC ( HUDKit.fla 1 centerText ) ,
,centerText ,
```



5 centerTextMC

```

centerTextMC      centerText      ,      textField
      centerTextMC      "on5"(
5      )      "on"(      3      )      1
      ,      ,centerTextMC      Alpha      0,      "on"      "on5"
TextFieldMC.SetText(RespawnText); // Set the text field of the Center Text.
CenterTextMC.GotoAndPlay("on"); // Play the fade-in animation, it will fade on its
                                own after ~3 sec.

      ,      centerTextMC      textField      ,      GotoAndPlay("on")
      Classic Tween      ~3

```

4.8 彈出消息和通知

```

popup      (      HUDKit fla      1      popup(      )      )      ,
      3      (      )      ,      popup      ,      3-5
      C++

```



6 popup

```

Popup      Classic Tweens      popup
      popup      4      :actions      popupNumber      popupText      popupBG      popupBG      "      "
      popupText      popupNumber      textField

      "on"      ,      GotoAndPlay()      ,
      ,      1      ,      Alpha      0

```

```

// If a KillStreak event is fired, display the kill streak pop-up
case FxHUDEvent::EVT_KillStreak:
{
    FxHUDKillStreakEvent* peventKS = (FxHUDKillStreakEvent*)pevent;

    // Format the number of kills
    String text;
    Format(text, "{0}", peventKS->GetSrcKillStreak());
    PTextFieldMC.SetText(text); // 设置 PopUp 的文本框。

```

```

        // Play the fade-in animation, it will fade on its own after ~4 sec.
        PopupMC.GotoAndPlay("on");
    }
    break;

```

```

        ,FxHUDEvent      KillStreak
    C++      popupText's textField      PTextFieldMC      ,      GotoAndPlay("on")
    popup

```

4.9 消息和事件日誌

```

log      (      HUDKit fla      Scene 1      log(      )      )      HUD      ,
      (
      ~3
      ,

```



13 log logMessage

```

    logMessage
    logMessage      ,      3
    C++      ,      htmlText      textField(      )

```

```

        FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip(),
LogMessage MovieClip      Log      ,

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()
{
    if (MessageMCs.IsEmpty())
    {
        // Request a new line in the SWF
        Value logline;
        LogMC.AttachMovie(&logline, "LogMessage", "logMessage"+NumMessages);
        FxHUDMessage::Movieclip* pmc = new FxHUDMessage::Movieclip(logline);
        MessageMCs.PushBack(pmc); // Add new Message MC to list of unused

```

```

        message movieClips.
    }
    FxHUDMessage::Movieclip* pmc = MessageMCs.GetFirst(); // Use an unused Message
                                                           movieClip.

    pmc->SetVisible(true);
    MessageMCs.Remove(pmc);
    return pmc;
}

```

```

        C++          C++          ,          Level Up          ,
    "You are now a [Rank Icon] [Rank Name]"(          [          ] [          ])          htmlText

```

```

void    FxHUDLog::Process(FxHUDEvent *pevent)
{
    String msg;
    switch (pevent->GetType())
    {
        case FxHUDEvent::EVT_LevelUp:
        {
            FxHUDLevelUpEvent* e = (FxHUDLevelUpEvent*)pevent;
            Format(msg, "You are now a <img src='rank{1}' /> {0}!",
                e->GetNewRankName(), e->GetNewRank());
        }
        break;
    }
}

```

```

FxHUDMessage::Movieclip* pmc = GetUnusedMessageMovieclip();
FxHUDMessage* m = new FxHUDMessage(msg, pmc); // 3 秒钟
Log.PushBack(m);
NumMessages++;

```

```

        HUD Update
void    FxHUDLog::Update()
{
    SF_ASSERT(LogMC.IsDisplayObject());

    Double rowHeight = 30.f;

    // Tick the message lifetimes
    Double yoff = 0;
    FxHUDMessage* data = Log.GetFirst();
    while (!Log.IsNull(data))
    {
        FxHUDMessage* next = Log.GetNext(data);
        if (data->IsAlive())
        {
            // Layout mgmt and animation

```

```

        Value::DisplayInfo info;
        info.SetY(yoff);
        data->GetMovieclip()->GetRef().SetDisplayInfo(info);
    }
    data = next;
    yoff += rowHeight;
}

// Layout management and animation
Value::DisplayInfo info;
info.SetY(LogOriginalY - (NumMessages * rowHeight));
LogMC.SetDisplayInfo(info);

// Remove dead messages
data = Log.GetFirst();
while (!Log.IsNull(data))
{
    FxHUDMessage* next = Log.GetNext(data);
    if (!data->IsAlive())
    {
        Log.Remove(data);
        NumMessages--;

        FxHUDMessage::Movieclip* pmc = data->GetMovieclip();
        pmc->SetVisible(false);
        MessageMCs.PushBack(pmc);

        delete data;
    }
    data = next;
}
}

Log, FxHUDMessage, 0, FxHUDMessage
, Log, FxHUDMessage MovieClips, MessageMCs
FxHUDMessage, Gfx::Value::DisplayInfo.SetY() Gfx::Value::SetDisplayInfo()
MessageMC

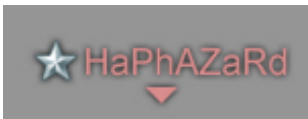
```

4.10 看板

```

billboard_container( HUDKit fla Scene 1 billboard )
                    3D, 3D
(Frustum),
,

```



7 billboard_enemy

UI 2D ,

Gfx::Value::SetDisplayInfo()

MovieClip (billboard_friendly, billboard_enemy)

billboard_label_(friendly/enemy)

billboard_label_*

textField billboard_friendly's "Show" Keyframe

billboard_friendly

Classic Tween

"Show"(), ,

C++ FxHUDView::BillboardCache::GetUnusedBillboardMovieclip :

```
if (UnusedBillboards.IsEmpty())
{
    Value::DisplayInfo info(false);

    Value billboard, temp;
    String instanceName;
    Format(instanceName, "{0}{1}", BillboardPrefix, BillboardCount++);
    CanvasMC.AttachMovie(&billboard, SymbolLinkage, instanceName);

    BillboardMC* pbb = new BillboardMC();
    pbb->Billboard = billboard;
    billboard.GetMember("label", &temp);
    temp.GetMember("textField", &temp);
    pbb->Textfield = temp;

    pbb->Billboard.SetDisplayInfo(info);
    UnusedBillboards.PushBack(pbb);
}
```

BillboardCache , CanvasMC ,
 BillboardCache billboard_enemy/billboard_friendly
 CanvasMC.AttachMovie() , ,

```

Billboard                                :UpdateBillboards() BeginProcessing() EndProcessing()
GetUnusedBillboardMovieclip()           BeginProcessing()
EndProcessing()

```

```

UpdateBillboards()

```

```

// Process friendlies
// Friendly billboards will always show names

```

```

BillboardMC* pbb = NULL;

```

```

FriendlyBillboards.BeginProcessing();
for (unsigned i=0; i < friendlies.GetSize(); i++)
{
    info.Clear();
    if (FriendlyBillboards.GetUnusedBillboardMovieclip(friendlies[i].pPlayer,
                                                         &pbb))
    {
        info.SetVisible(true);
        pbb->Billboard.GotoAndPlay("show");
        // Load player info
        String title;
        Format(title,
               "<img src='rank{0}' width='20' height='20' align='baseline'
               vspace='-7' /> {1}",
               friendlies[i].pPlayer->GetRank()+1,
               friendlies[i].pPlayer->GetName());
        pbb->Textfield.SetTextHTML(title);
    }
    info.SetX(friendlies[i].X);
    info.SetY(friendlies[i].Y);
    info.SetScale(friendlies[i].Scale, friendlies[i].Scale);
    pbb->Billboard.SetDisplayInfo(info);
}

```

```

FriendlyBillboards.EndProcessing();

```

```

UpdateBillboards()
GotoAndPlay("show") , Gfx::Value::SetDisplayInfo() MovieClip's X Y ,
Gfx::Value::SetText()

```

```

textField    htmlText
src='rank{0}' HTML , + 1 (friendlies[i].pPlayer->GetRank()+1)

```

```
img                                     vspace {1}  
, (friendlies[i].pPlayer->GetName())
```


5 迷你地圖視圖

(Minimap) C++ (Apps\Kits\HUD\):

- **FxMinimap.h**
- **FxMinimapView.h/.cpp**

5.1 迷你地圖視圖

:

- **FxMinimapIconType** , ID
- **FxMinimapIcon**
- **FxMinimapIconCache**

, / , MinimapIconFactoryBase

MovieClips
Update() ,

, : textField :

o *PlayerIcon* ,

o *FlagIcon*

o *ObjectiveIcon*

o *WaypointIcon*

- **FxMinimapView** UpdateView()

PlayerIcon::Update() (FxMinimapView.cpp; 116) Direct Access

API MovieClip MovieClip SetDisplayInfo
(Alpha), SetMember()
,Gfx::Value::SetMember Gfx::Movie::SetVariable :

```
virtual void Update(FxMinimapView* pview, FxMinimapEntity* pentity,
                    float distSquared)
{
    SF_UNUSED(distSquared);
    bool hasIcon = (pentity->GetIconRef() != NULL);
    PointF pt = pentity->GetPhysicalPosition();
    bool showDir = pview->IsShowingPlayerDir();
```

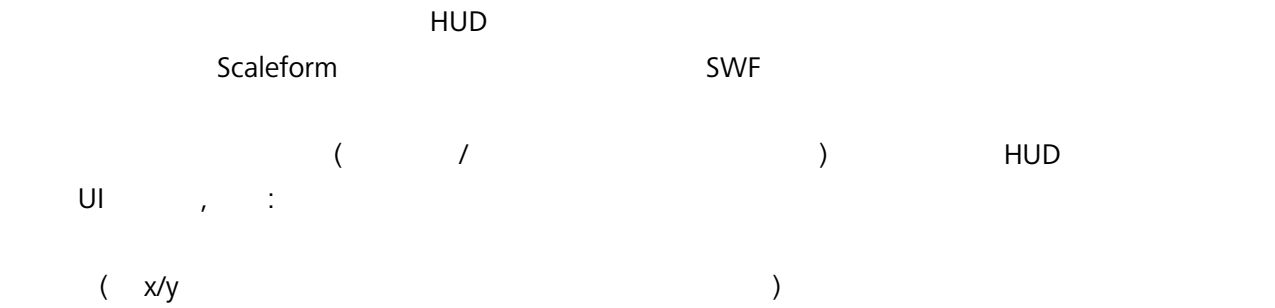
```

// If entity did not have an icon attached before, then wait for the
// next update to set the state data. picon is most probably not initialized
// (the movie needs to advance after creation)
if (hasIcon && (State != (int)showDir))
{
    // state 0: no arrow, state 1: show arrow
    Value frame(Double(showDir ? 2.0 : 1.0));
    MovieClip.Invoke("gotoAndStop", NULL, &frame, 1);
    State = (int)showDir;
}

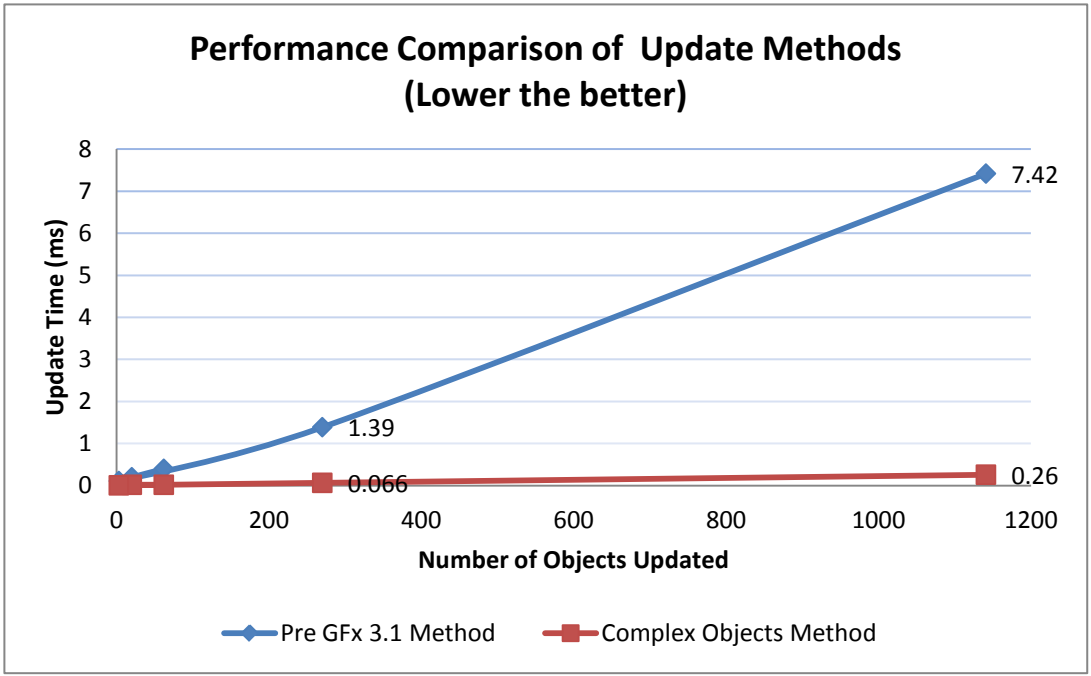
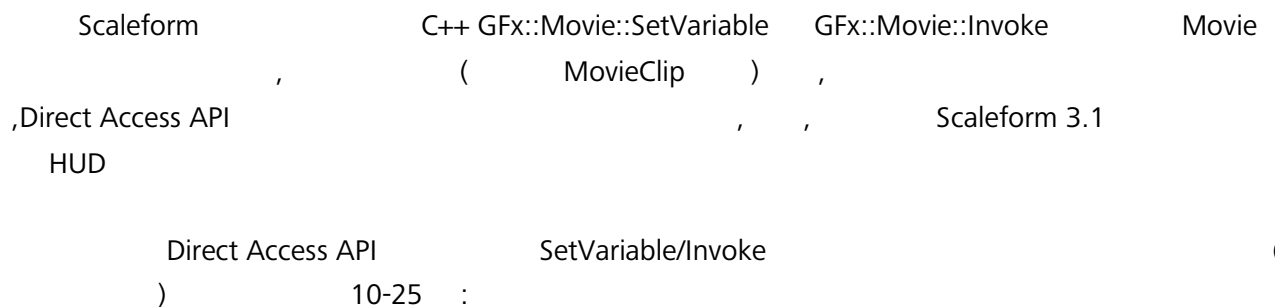
pt = pview->GetIconTransform().Transform(pt);
Value::DisplayInfo info;
if (State)
{
    info.SetRotation(pview->IsCompassLocked() ?
        (pentity->GetDirection() + 90) : (pentity->GetDirection() -
            pview->GetPlayerDirection()));
}
info.SetPosition(pt.x, pt.y);
MovieClip.SetDisplayInfo(info);
}

```

6 性能分析



6.1 性能統計資料



8 Direct Access API SetVariable/Invoke

,Direct Access , 1
 – HUD UI

Scaleform 15 HUD
 Flash HUD :

	FPS				
Windows Vista MacBook Pro	1148	0.018	0.512	0.017	0.527
Xbox 360	1136	0.032	0.454	0.010	0.497
PlayStation 3	752	0.044	0.668	0.021	0.733

1 Scaleform HUD

1 ,HUD , 1 –
 HUD UI

6.2 記憶體明細表

HUD GFxExport

1. .GFX
GFx

1858kb

- HUDKit.gfx - 33kb
- Minimap.gfx - 50kb
- fonts_en.gfx - 46kb
- gfxfontlib.gfx - 100kb

- HUDKit.gfx
 - o UI - 603kb
 - o - 72kb
 - o - 51kb
 - HUD , 43kb
- Minimap.gfx
 - o - 644kb
- Map.jpg - 259kb